

Your company has a two-tier application with a web-based frontend for interacting with users and an RDBMS backend that stores user data. You need to find a way to gradually move your systems from on-premises to a cloud deployment. How should you implement a tiered hybrid cloud deployment?

- A) Use a private cloud for performing changes to frontends

✖
- B) Deploy frontend applications to a public cloud

✔
- C) Store data that is restricted by regulations on a public cloud

✖
- D) Migrate backend applications to a public cloud

✖

Explanation

You should deploy frontend applications to a public cloud. Frontend applications require frequent changes and updates as they face users and are performance centric. By moving frontend applications to a public cloud, you can have an automated continuous integration and continuous delivery (CI/CD) system for efficient deployment of updates to applications.

You would not migrate backend applications to a public cloud. Backend applications can have various system dependencies that are very complex, and the task of migrating these applications to a public cloud can be an involved and challenging process. By focusing on moving the frontend applications to the cloud first, you can perform cloud integration faster and utilize cloud-native services for increased availability and resiliency for the system.

You would not store data that is restricted by regulations on a public cloud. Any data that is subject to regulatory restrictions should ideally be stored in an on-premises environment. This approach ensures greater security of the data.

You would not use a private cloud for performing changes to frontends. This is because a public cloud system offers features like CI/CD for rapid and automated development and updates for applications. Additionally, cloud services, including Cloud Internet of Things (IoT) and Cloud Content Delivery Network (CDN), can be useful for creating user interfaces, data ingestion from IoT sources, and creating application programming interfaces (APIs).

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

You have several teams working on creating applications, and you need to design a system that enables the applications made by each team to interact with each other asynchronously. Which of these services should you utilize for this scenario?

- A) Pub/Sub

✔
- B) BigQuery

✖
- C) Dataprep

✖
- D) Dataproc

✖

Explanation

You should utilize Pub/Sub for this scenario. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture. Pub/Sub is useful for application developers that need to create a communication system between applications that are built independently of each other.

You would not utilize Dataproc in this scenario. Dataproc is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not utilize BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. It allows you to perform queries on petabytes of data within minutes.

You would not utilize Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of ML and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Pub/Sub](#)

Related Material

Your company has recently moved to a cloud deployment and is struggling to manage its growing cloud costs. You are asked to suggest some ways to get costs under control. Which of these should you suggest? (Choose two.)

- A) Set usage quotas for resources

✔
- B) Store all data with the standard class

✖
- C) Run all VMs continuously

✖
- D) Create alerts for budget overruns

✔
- E) Duplicate data across buckets

✖

Explanation

You should suggest the company set usage quotas for resources and create alerts for budget overruns. Some controls available to companies for keeping cloud costs and usage under control include budgetary alerts, usage quotas, and company permissions. Setting resource quotas and performing throttling must ensure that cost overruns do not occur due to resource utilization exceeding the specified budgets. Using Google Cloud tools, you can perform automated throttling, capping of resource consumption, and broadcast budget notification alerts on company communication channels, such as Slack.

You would not store all data with the standard class. This action can cause wasted capital due to overspending on storage. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

You would not run all virtual machines (VMs) continuously. Using auto start and stop for VMs can help reduce Compute Engine's operating costs. By deactivating VMs during non-business hours rather than allowing VMs to run continuously, you can gain significant cost savings. You can use a serverless system to automate the process of VM shutdowns by using Cloud Pub/Sub, Cloud Scheduler, and Cloud Functions.

You would not duplicate data across buckets. Duplicating data causes a waste of storage and capital. Some ways you can prevent creating duplicate data is by using multi-region buckets and setting object versioning on Cloud Storage correctly. By specifying an object versioning policy, you can maintain a suitable number of copies of data, and you can also use the bucket lock version to protect from accidental deletion of objects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #4 of 60

Question ID: 1495297

You need to store the events generated by various Google Cloud services in buckets that need to be localized to specific regions for regulatory and compliance requirements. Which Google Cloud service would be the most appropriate choice for this?

- A)** Cloud Profiler
- B)** Cloud Endpoints
- C) Cloud Logging**
- D)** Cloud Trace
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Explanation

You would use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) for collecting log entries, which are passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or import.

You would not use Cloud Profiler for this scenario. Cloud Profiler is used to find and remove any possible performance issues from your applications currently in production. This is accomplished by Cloud Profiler regularly profiling the consumption of resources across your production applications. Cloud Profiler creates an analysis of the performance of an application by collecting data related to CPU usage and memory allocation.

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance.

You would not use Cloud Endpoints for this scenario. Cloud Endpoints provide a system for managing APIs that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #5 of 60

Question ID: 1497147

You are a cloud services consultant for a digital publishing company that provides the news by covering the latest world events through web and mobile apps. You want to find a way for the news to be made available in several local languages across the world automatically. Which of these services can you use for this?

- A)** Cloud Vision
- B) Cloud Natural Language API**
- C)** Cloud Translation
- D)** Cloud Spanner
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Explanation

You would use Cloud Translation for this scenario. Cloud Translation or Translation Artificial Intelligence (AI) performs translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers for understanding image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

You would not use the Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language AI enables you to gain an understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be utilized by developers either through API calls or through reference to a document that may be stored on Cloud Storage.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner to provision a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Translation AI](#)

Related Material

Question #6 of 60

Question ID: 1497405

You want to ensure maximum security of your sensitive data and also leverage cloud-native services like serverless computing. Which cloud deployment should you opt for?

- A) Hybrid Cloud
- B) Multi-Cloud
- C) Private Cloud
- D) Public Cloud

Explanation

You should opt for a hybrid cloud infrastructure model. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

You would not opt for a private cloud infrastructure. In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This deployment option does not include systems that are a part of a public cloud service offering.

You would not opt for a public cloud infrastructure. In this deployment, the infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This deployment option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

You would not opt for a multi-cloud infrastructure. In this deployment, the data is still stored on a public system, which can cause security concerns for the company. A multi-cloud deployment involves using multiple public CSPs where each public cloud provides a different service for the company. This allows a company to use one public cloud for database services and another one for analytics. This deployment option lets a company use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #7 of 60

Question ID: 1495179

Your company is looking to build a new e-commerce portal for its customers and wants to take the appropriate cloud adoption approach. Which would be a suitable option for this requirement?

- A) Use the cloud to only modernize the infrastructure
- B) First migrate the applications and then rearchitect them on the cloud
- C) Create a new infrastructure and application entirely on the cloud
- D) Rearchitect applications first to make them ready for the cloud

Explanation

You will create a new infrastructure and application entirely on the cloud. This approach is also called “Invent” or “Greenfield”. To make this happen, a company needs to have access to a wide range of development skills and strong sponsorship from executives. This approach features the creation of Cloud Centers of Excellence (CCEs) which are teams comprised of people from different backgrounds who perform experimentation and create documentation for new methods of scaling and working on the cloud with a set of key best practices.

First migrating the applications and then rearchitecting them on the cloud is not a suitable approach for this scenario. This approach is also called “Move first and then change” or “Lift and shift”. This approach is better suited for a company that wants to adopt a conservative approach to cloud adoption. This method features fewer disruptions to existing work processes and operations. In this approach, the changes for cloud adoption take more time and gain momentum gradually as new work methods are accepted by more people in the company.

Rearchitecting applications first to make them ready for the cloud is not a suitable approach in this scenario. This is also called “Change first before moving” or “Improve and move”. This approach is suitable for companies that take a forceful approach for cloud adoption and features the use of a DevOps and automation-based approach to work as opposed to the traditional waterfall delivery model.

Using the cloud to only modernize the infrastructure is not suitable for this scenario. This approach is also simply called “Move”. This approach is for companies that just want to modernize their infrastructure layer to avail of virtualized infrastructure for disaster recovery (DR) and big data analysis. Big data can include structured data from databases and unstructured data from Internet of Things (IoT) sensors, images, emails, and social media.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > Managing Change in the Cloud](#)

Related Material

Question #8 of 60

Question ID: 1495222

The company you work for wants to find more secure and cost-effective methods of managing its systems. You are looking to select the appropriate cloud deployment for this. Which of these deployment configurations uses a hybrid cloud infrastructure?

- A) Using an on-premises data center for storing raw data along with a public cloud for data analytics**
- B) Storing all organizational data and applications in an on-premises data center**
- C) Using SaaS from one cloud provider and PaaS from another**
- D) Using one public cloud service provider for a web frontend and a different one for the backend**

Explanation

Using an on-premises data center for storing raw data along with a public cloud for data analytics is an example of a hybrid cloud infrastructure. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

Using one public cloud service provider (CSP) for a web frontend and a different one for the backend is not an example of a hybrid cloud deployment but of a multi-cloud deployment. A multi-cloud deployment involves using multiple public CSPs where each public cloud services a different function for the company. A company can then use one public cloud for database services and another one for analytics. This lets a company use its preferred cloud provider for specific services. Remember that using multiple public cloud service providers indicates a multi-cloud infrastructure.

Using Software as a Service (SaaS) from one cloud provider and Platform as a Service (PaaS) from another is not an example of a hybrid cloud infrastructure but of a multi-cloud infrastructure. This is because both SaaS and PaaS are public cloud offerings, and using several public clouds is a feature of multi-cloud deployments. The PaaS model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary operating system and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and operating system(s) to make it happen. The SaaS model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including mobile devices and web browsers. An example of this is email accessible from a web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it.

Storing all organizational data and applications in an on-premises data center is not an example of a hybrid cloud infrastructure but of a private cloud infrastructure. In this infrastructure, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premise data center.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #9 of 60

Question ID: 1495238

You are creating a plan for digitally transforming workloads for your company. What would be the purpose of using cloud-native applications during application modernization?

- A) Consolidating a single unified backend for redundancy**
- B) Avoiding cost overruns caused by application decoupling**
- C) Partitioning workloads for failure protection**
- D) Using APIs for ensuring all business teams work at the same pace**

Explanation

Using cloud-native applications allows for partitioning workloads for failure protection. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services as well as protection from failures by maintaining several read replicas of a service within a system that is distributed.

Consolidating a single unified backend for redundancy is not a purpose of application modernization. You would instead spread backends and other services over several availability zones (AZs) and use read replicas and backups to ensure that the system is not affected by failures. You may need to maintain certain services in a specific geographical location for compliance, and you may need to distribute other services globally for low latency content distribution to users.

Avoiding cost overruns caused by application decoupling is not a purpose of application modernization. Performing decoupling of applications as part of application modernization ensures that services that scale up or down more frequently are separated from other services that are more stable. This way, you cut back on costs associated with scaling up of services that are not required or are unrelated.

Using application programming interfaces (APIs) to ensure all business teams work at the same pace is not a purpose of application modernization. You would use APIs to perform application decoupling for existing business services. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #10 of 60

Question ID: 1497137

You want to build greater safety and accountability into your company's AI products. To help you achieve this, you are looking for relevant tools that are part of Google's Responsible AI framework. Which tools would you use? (Choose two.)

- A) Cloud Storage

✖
- B) Looker

✖
- C) Cloud Trace

✖
- D) TensorFlow open-source toolkit

✔
- E) Model cards

✔

Explanation

Model cards, TensorFlow open-source toolkit, and Explainable artificial intelligence (AI) are tools used as a part of Google's Responsible AI framework. Responsible AI tools allow you to inspect and analyze existing AI models to ensure safer products, increased customer trust, and ethically sound innovation. Explainable AI is a collection of frameworks and tools for understanding and interpreting machine learning (ML) model predictions and is integrated with various Google services. Model cards provide a structured way to organize ML Models' key facts, limitations, and capabilities. TensorFlow is an open-source platform used for ML. TensorFlow can be utilized for integrating the processes required for Responsible AI into an existing workflow for ML.

Cloud Trace is not used as a part of the Responsible AI system, but it is used for tracing purposes. Cloud Trace gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Cloud Storage is not used as a part of the Responsible AI system. It allows you to store and access data on Google Cloud using a Representational State Transfer (RESTful) service. It also provides features for security and data sharing. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Looker is not used as a part of the Responsible AI system. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive which allow for analyzing data in real-time through a trusted data model.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Responsible AI](#)

Related Material

Question #11 of 60

Question ID: 1495269

Your company is planning to perform a digital transformation using Google Cloud. Before the transformation process begins, the company needs to know more about Google's security posture for its platform. How does Google implement security for data stored on its cloud infrastructure?

- A) Multi-party authorization

✖
- B) Encryption at rest

✔
- C) Virtual network traffic encryption

✖
- D) Encrypting inter-service communication

✖

Explanation

Google implements encryption at rest to secure data stored on the cloud infrastructure. This is done by utilizing multiple layers of encryption for the protection of data at rest. Google’s storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner as well as a key management service. Cloud Key Management Service (KMS) is a Google Cloud service that customers can use to manage encryption keys. A related concept is encryption in transit, where Google protects data that is transferred between an on-premises system and Google Cloud or between services. This is done by encrypting the data before it is sent, performing authentication for endpoints, and then decrypting and verifying the data upon arrival at the destination.

Multi-party authorization is not related to the security of stored data but is a way of managing access between services. Using a multi-party authorization system, you can implement a two-person restriction that requires an engineer to get approval from another authorized engineer before performing an action that may be of a sensitive nature. Using this system allows for massive scalability of access management systems for a large number of services that operate on the cloud infrastructure.

Virtual network traffic encryption is not related to the security of stored data, but it is a way of encrypting communication between services. Google encrypts all of the traffic on its virtual networking systems. Additionally, there is authentication performed between services running in the infrastructure, as well as encryption of inter-service data transfer. This layered security approach ensures the security of data even if the network is breached or there are compromises of network devices.

Encrypting inter-service communication is not related to the security of stored data but is a way of ensuring that a service is configured to only receive Remote Procedure Call (RPC) requests from an authorized service. An example is, configuring Google Contacts to allow RPC requests from Gmail. This involves Gmail presenting an end-user permission ticket inside of an RPC request. The ticket ensures the Google Contacts returns data only for the user mentioned in the ticket, which creates a safeguard.

Some fundamental cloud security terms you should know include:

- Privacy – This is ensuring that data that belongs to the company is not shared or leaked to any unauthorized entity inside or outside the company. Privacy controls include access management and authentication systems.
- Availability – This is the ability of a system to offer services as measured as a ratio between total uptime versus downtime. For example, the five 9s of availability indicate that a system is available 99.999% of the time.
- Security – This includes all the technologies and processes in place to ensure that the company’s IT systems are safe from unauthorized access or damage. Security systems can range from intrusion detection and prevention systems to multi-factor authentication (MFA) systems for securing access.
- Control – This includes all the systems in place that ensure resources are only accessed by authorized users and in an appropriate manner. Control systems include landing zones that ensure compliance as well as resource policies that ensure safe access to resources.
- Zero Trust – This is a security strategy for cybersecurity where policies for security are enforced using strong user authentication and the principle of least privilege. Zero Trust means that no entity should be trusted, and each actor on the system needs to be verified. Zero Trust systems ensure an improved customer experience, simple network infrastructures, and better defense against cyberthreats.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google Infrastructure Security Design Overview](#)

Related Material

Question #12 of 60

Question ID: 1495285

A software development company is using Google Cloud for its development environment. It is now considering using DevOps. What is the purpose of incorporating DevOps into a production environment? (Choose two.)

- A) Increase in-person processes

✗
- B) Increase software stability

✓
- C) Make deployments quicker

✓
- D) Perform less frequent testing

✗
- E) Lessen the need for monitoring tools

✗

Explanation

Increasing software stability and making deployments quicker are both goals of using DevOps. DevOps is an initiative that is geared toward increasing the speed of delivering software, making services more reliable, and creating ownership that is shared among all stakeholders. DevOps principles are used to enforce security best practices from the beginning of each development project. Google’s DevOps Research and Assessment (DORA) team has found certain key capabilities that an organization can leverage for improved performance and faster software delivery. These capabilities are related to the company’s culture, measurement systems, technology, and processes.

Increasing in-person processes is not the purpose of DevOps. Using automation is a key feature of the DevOps philosophy, which seeks to create an automated model for application development that results in more error-free operations. The automated approach also helps ensure that changes can be rolled back, if required, and system stability is ensured using safe configurations.

Performing less frequent testing is not the purpose of DevOps. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used for making changes to ensure system stability and avoid configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually. Continuous testing (CT) is a part of CD, which ensures testing occurs regularly and in sync with delivery.

Lessening the need for monitoring tools is not the purpose of DevOps. The system health and reliability need to be monitored and measured as part of the DevOps philosophy. For this, various monitoring tools need to be used. Monitoring helps understand the performance of applications and map resource consumption to specific code areas of the application for further optimization.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Cloud.google.com > DevOps](#)

Related Material

Question #13 of 60

Question ID: 1497151

A company’s cloud deployment has already had a lot of budget overruns, and the company needs to cut back on cloud expenditure immediately. You need to implement financial governance strategies to gain more control over cloud costs. How would you achieve this?

- A) Decommissioning unused on-premises servers

✗
- B) Ensuring each user can provision cloud resources

✗
- C) Using organizational policies for resources

✓
- D) Providing cost viewing to all cloud users

✗

Explanation

You would use organizational policies for resources to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources. You will also use cost management controls to ensure you don't spend beyond your stipulated budget and plan costs properly. You can set limitations on cost accumulations using quotas and perform monitoring for usage and costs using budgets. You can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Providing cost viewing to all cloud users is not the correct way to gain more control over your cloud costs because each user on a cloud deployment cannot be allowed more privileges than their job role requires, as this can cause security as well as cost management issues. Cloud resource usage and cost viewing permissions need to be made based on the specific role a user performs in the organization.

Ensuring each user can provision cloud resources is not the correct way to gain more control over your cloud costs because allowing all users to provision cloud resources can cause major cost overruns and cost control issues. Correct financial governance on the cloud is to allow resource provisioning based on the user's role and position in the organizational hierarchy.

Decommissioning unused on-premises servers is not the correct way to gain more control over your cloud costs because the state of on-premises infrastructure does not affect cloud operating costs as the two systems are disparate. Decommissioning unused servers from an on-premises data center can help improve the operational effectiveness of an on-premises data center; However, this will not have an effect on how resources are provisioned on the cloud deployment, which requires a different system be in place to ensure cost control, such as using IAM policies.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #14 of 60

Question ID: 1495940

Your company has multiple teams that have hundreds of employees in them. You need a centralized way to manage user access to cloud resources. Which of these systems would you need to use to authorize which users can access specific Google Cloud resources?

- A) Cloud IAM**
- B) Secret Manager**
- C) Encryption at rest**
- D) Billing APIs**
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Explanation

You would use Cloud Identity and Access Management (IAM) for this scenario. Cloud IAM enables you to manage cloud resources centrally with granular access control and visibility. With IAM, you can grant access at the resource level, such as providing the role of Subscriber to a user for a Pub/Sub topic. This way, IAM allows you to grant roles in addition to Owner, Editor, and Viewer roles. IAM also provides Recommender that can enable you to detect overly permissive access policies and rightsize them. With IAM, resource access is controlled using contextual attributes, such as IP address or resource type. You can perform IAM policy management and creation using the Google Cloud Console, programmatically using IAM methods, or by command-line access using the gcloud tool. Google Accounts are also supported by IAM. You can utilize Cloud Identity with IAM to synchronize user accounts across multiple projects. IAM also provides workforce identity federation that enables you to perform authentication and authorization of a workforce using external identity providers (IdPs). IAM can provide a full audit history that includes granting, removal, and delegation of permissions for compliance needs.

You would not use Secret Manager for this scenario. You use Secret Manager for storing sensitive data like passwords, application programming interface (API) keys, and certificates securely. With Secret Manager, you have a single source of truth for accessing, managing, and auditing secrets on a Google Cloud deployment. Secret Manager provides lifecycle management with versioning for pinning requests to the newest versions of secrets, as well as the ability to use Cloud Functions to automate the process of rotating secrets. It also allows you to select regions for the replication of your secrets data. Secret Manager is integrated with Cloud IAM for controlling access to secrets.

You would not use encryption at rest in this scenario. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner, as well as a key management service.

You would not use billing APIs for this scenario. You can use billing APIs to access various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. This is a way of ensuring clarity for financial governance on a cloud deployment. Other methods include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Cloud Identity and Access Management \(IAM\)](#)

Related Material

Question #15 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

- A) Dataproc**
- B) Dataflow**
- C) Looker**
- D) Cloud Trace**
-
-
-
-

Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Related Material

Question #16 of 60

Question ID: 1497124

Which of these are characterized as key advantages of moving to a data-driven culture? (Choose two.)

- A) Use of a Zero Trust security model for data sharing**
- B) Less reliance on fully managed services for data management
- C) Remove the need for data governance
- D) Creation of duplicate data for use within an organization
- E) Easy access to analytics data for the business users of a company**

Explanation

Easy access to analytics data for the business users of a company and the use of a Zero Trust security model for data sharing are both key advantages of moving to a data-driven culture. A data-driven culture gives data the topmost priority and forms the basis of its decision making for its strategies and teams. A data-driven culture incorporates people, processes, and technology, and ensures that a business:

- operates with trust
- ensures insights are democratized
- increases its agility
- applies business intelligence

Less reliance on fully managed services for data management is not an advantage of moving to a data-driven culture. This is because data can be used efficiently when using more managed integrated systems. This bridges any skill gaps that may exist for users and enables them to operate, scale, and manage data with high levels of agility. This allows for greater user collaboration and removes bottlenecks and silos in an organization.

The creation of duplicate data for use within an organization is not an advantage of moving to a data-driven culture. A data-drive culture seeks to create a single source of reliable data within an organization that can be accessed by users when they need it. This can be achieved by ingestion of data into a data warehouse and building data models with business intelligence systems, such as Looker. This makes relevant data available for all teams without issues related to redundancy and provisioning overhead.

Removing the need for data governance is not an advantage of moving to a data-driven culture. Data governance is an integral part of a data-driven culture and ensures that only the correct users can access relevant data and also ensures that data is safe throughout its lifecycle.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material

Question #17 of 60

Question ID: 1495194

- A global banking company needs a cloud-based managed service for its transcontinental ATM transactions and online banking services that execute a large number of real-time database transactions using the internet. Due to the nature of the transactions, the service needs to be high performance and have strong consistency for the transactions. Which service would be the right choice to meet these requirements?
- A) Cloud Data Fusion**
- B) Cloud Storage
- C) Cloud Spanner**
- D) Cloud Trace

Explanation

Cloud Spanner is the correct choice for this requirement because it is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner to provision a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability, up to 99.999%, and strong consistency worldwide. Cloud Spanner ensures zero scheduled downtimes and allows for schema changes to be done online. Cloud Spanner has unlimited scalability, which can match the organization’s growth and ensure that database transactions are high-performance. It also provides auto-sharding, which is done based on the size of the data and a load of requests. Sharding is the process of dividing a larger database into smaller portions called “shards” for better speed and manageability.

Cloud Storage is not the appropriate choice for this scenario because you would use Cloud Storage for unlimited storage and retrieval of objects on the cloud. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

Cloud Trace is not the appropriate choice for this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Cloud Data Fusion is not the appropriate choice in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for portability of data pipelines and integration for multi-cloud and hybrid cloud deployments, which enables users to remove existing solos and access previously inaccessible data.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Spanner](#)

Related Material

Question #18 of 60

Question ID: 1497141

Your development team has limited experience building intelligent systems for image classification based on predefined labels. You need a way to train customized models with the least possible effort so that workloads can be parallelized and less time is spent on manual tasks. Which of these technologies can help you achieve this goal?

- A) AutoML

B) Looker

C) Dataproc

D) Cloud Spanner
- ✔

✖

✖

✖

Explanation

You can use Automated Machine Learning (AutoML) to achieve the necessary objective. AutoML enables you to perform training for high-grade machine learning (ML) models with the least manual effort and knowledge of ML. A user can essentially build a working ML model within a few minutes. Developers can submit their datasets and get to use trained ML models that are built using Google technologies, such as transfer learning and Neural Architecture Search. Neural Architecture Search is Google's technology that provides a way to find, generate, evaluate, and train multiple neural architectures for choosing a solution for a customer automatically. Some key technologies within AutoML include:

- AutoML Natural Language – This allows users to perform categorization of input text into their own predefined labels.
- AutoML Tables – This allows data scientists and developers to build and deploy ML models for structured data at high speeds and scale.
- AutoML Translation – This allows developers to use Google Neural Machine Translation (GNMT) for specific use cases for translation.
- AutoML Video – This allows developers to create ML models for video that are customized and scalable.
- AutoML Vision – This allows developers to train ML models for performing image classification based on custom labels.

The underlying concept behind AutoML involves searching using special algorithms to find solutions that are most optimal for each part of an existing pipeline for ML. This enables a system to be built that automates three main aspects of ML including feature engineering, neural architecture search, and hyperparameter optimization.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for provisioning a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > AutoML](#)

Related Material

Question #19 of 60

Question ID: 1497153

You are a consultant for a stock trading company that is setting up its Google Cloud environment. You need to guide the team on some key issues that the company needs to be aware of when managing cloud environments. Which of these concerns should you specify? (Choose two.)

- A) Less cloud usage transparency

B) Lack of cost predictability

C) Lack of on-premises servers

D) More overhead using cloud services

E) Security of servers housing virtual machines
- ✔

✔

✖

✖

✖

Explanation

You should specify lack of cost predictability and less cloud usage transparency as key issues the company should be aware of. Lack of cost predictability can be an issue for companies managing cloud environments. A customer application may experience surges in user traffic alternated with periods of low levels of traffic. Due to the unpredictable nature of application usage, it can become difficult for a cloud administrator to specify resource limits. Setting resource limits too low can lead to application performance issues during peak usage, and provisioning more resources than required can lead to wasted capital. Less visibility and transparency in cloud usage leads to the wastage of cloud expenditure due to unutilized resources. By using cloud monitoring tools, a cloud consumer can identify resources that are unused and deprovision them so that they do not contribute to the overall billing. This helps reduce unnecessary costs for a cloud deployment.

The lack of on-premises servers is not a concern when managing a cloud environment because the set-up of an on-premises data center and its operation are not related to a cloud environment. To increase the computing power of a cloud deployment, you can perform scaling either manually or automatically, which provisions more virtual servers when required.

More overhead using cloud services is not a concern when managing a cloud environment. By using fully managed services on Google Cloud, you can reduce your operating costs by removing all the overhead that would be required for various tasks, which include resource provisioning, backups, updates, and patching. This reduction in costs allows you to better utilize your capital to grow your business.

The security of servers housing virtual machines (VMs) is not a concern when managing a cloud environment. This is the responsibility of the cloud service provider (CSP). As per the shared responsibility model of cloud environments, the customer is responsible for securing the data and applications that are hosted on the cloud, and the CSP secures the entire cloud infrastructure, which includes networking systems and physical servers that house VMs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com >A guide to financial governance in the cloud](#)

Related Material

Question #20 of 60

Question ID: 1497146

Your company provides recruitment consultant services for a wide user base and needs a technology solution for classifying and extracting structured data from inside user application forms. Which of these technologies would you use for this requirement?

- A) Cloud Translation

✖
- B) Visual Inspection AI

✖
- C) Document AI

✔
- D) Media Translation API

✖

Explanation

Document Artificial Intelligence (AI) is the appropriate choice for this scenario. You would use Document AI for classification and extraction of structured data from existing documents. This can enable you to find key insights and perform automation of organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information that is extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for the automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gPRC) framework.

Visual Inspection AI is not the appropriate choice for this scenario. You would use Visual Inspection AI for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables the development of advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Document AI](#)

Related Material

Question #21 of 60

Question ID: 1495271

Your company is evaluating the key security benefits of using Google Cloud for its operations. How does Google Cloud allow you to implement preventative controls for ensuring the security of your cloud deployment?

- A) Performing logging and monitoring

✖
- B) Using third-party libraries

✖
- C) Setting budgetary alerts

✖
- D) Setting IAM permissions

✔

Explanation

Setting Identity and Access Management (IAM) permissions is an example of using preventative controls to secure a cloud deployment. You can implement preventative controls using security policies and system architecture. A security policy includes constraints that are implemented programmatically to protect a company and its cloud resources from unauthorized access. The system architecture includes the protection that is available using the existing infrastructure. Preventative controls can include routing configuration, firewalls, subnetting, virtual private clouds (VPCs), and resource hierarchies.

Using third-party libraries is not an example of preventative controls. Third-party libraries are used to provide various functionalities that a development team may require for a project. However, third-party libraries need to be validated for the presence of security issues or bugs that need to be patched. You need to assess how the library sits in with the rest of the development stack. If the library is used for authentication, then it needs more scrutiny than if it is used for the User Interface (UI). Some general guidelines for the use of third-party libraries are to first analyze the scale and usage of the library amongst users globally and gauge their experiences. You also need to know how well the library is managed so future risks can be alleviated.

Performing logging and monitoring is not an example of preventative controls. These are both examples of detective controls. Alerting, analysis, monitoring, and logging are used to identify any malicious activity inside an organization. Integrating Security Information and Event Management (SIEM) systems with Google Cloud Logging is how detective controls can be implemented.

Setting budgetary alerts is not an example of preventative controls. These are controls available to companies for keeping cloud costs and usage under control. These include budgetary alerts, usage quotas, and company permissions. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide > Google Cloud foundation security model](#)

Related Material

Question #22 of 60

Question ID: 1495168

Who is responsible for security in an IaaS cloud service model?

- A) The ISP used by the customer

B) Both the customer and cloud service provider

C) The customer only

D) The cloud service provider only
- ✖

✔

✖

✖

Explanation

In an Infrastructure as a Service (IaaS) cloud service model, both the customer and cloud service provider (CSP) share the responsibility of security for the cloud deployment. Neither the customer nor the CSP take complete responsibility for the security of the cloud deployment. In a shared responsibility model, the customer is responsible for security in the cloud and the CSP is responsible for security of the cloud. The CSP's responsibility includes the security of the data center, physical servers, and services. The customer's responsibility includes the security of the workloads. To summarize, the following are the responsibility of the CSP:

- Data center security
- Virtualization infrastructure
- Software and hardware infrastructure
- Networking infrastructure
- Global infrastructure, including Availability Zones, Regions, and Edge locations

Customers are responsible for securing all the data that they create and store in the cloud. The following are the responsibility of the customer:

- Patching and configuration of operating systems running on virtual machine instances
- Setup of security groups
- Administration of user accounts and permissions

The Internet service provider (ISP) that provides Internet services to the customer is not responsible for security of a cloud deployment. The ISP only provides the necessary Internet connectivity for access to cloud resources but is not involved with securing either the underlying cloud infrastructure or of applications and data stored on a cloud. Additionally, when connecting to a cloud account over the Internet, security mechanisms like virtual private networks (VPNs) may need to be used for securing data transfer.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

Related Material

Question #23 of 60

Question ID: 1495903

A company is currently considering cloud adoption with its workloads operating individually without a collective strategy for future scalability. The company wants to keep its costs of operation as low as it can and move to the cloud with minimal possible interruptions to its operations. Based on this data, which cloud maturity phase of the Cloud Adoption Framework would you place the company at?

- A) Transformational

B) Tactical

C) Strategic

D) Scale
- ✖

✔

✖

✖

Explanation

Based on the assessment, the company is currently at the tactical phase of the Google Cloud Adoption Framework (CAF) which is characterized by a need to minimize current operational costs and make minimal changes to existing IT teams, technology, and processes. An organization in a tactical phase can seek to optimize costs by making better use of compute and storage resources that are currently underutilized. It can also seek to remove the delays and overhead associated with manual procurement and provisioning of resources.

The company is not at the strategic phase of the CAF. The strategic phase is characterized by efforts to modernize the architectures of a company's IT solution by using cloud-native services. This phase also includes some changes made to the existing IT teams, technology, and processes.

The company is not at the transformational phase of the CAF. The transformational phase is characterized by efforts to fully reorganize the IT organization of the company and enable more knowledge sharing to ensure individuals and teams can make more decisions with autonomy. This phase has a cloud-first approach by seeking to leverage the best cloud services and best practices as per current norms. The transformational phase helps meet a long-term objective for a business to convert its IT systems into an engine of innovation that enables business transformation.

Scale is not a phase of the CAF but is a theme for cloud adoption. There are four cloud adoption themes as per the CAF which include learn, lead, scale, and secure. The learn theme measures the learning systems in place for an organization to upskill its IT workforce. The lead theme measures the level of support that the IT workforce receives from the company's leadership for cloud migration. The scale theme measures the depth of usage of services that are native to the cloud for reducing overhead in operations as well as the level of automation you perform for policies and processes which were operated manually. The secure theme measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric.

The three maturity phases component of the CAF gauge the readiness of a company for successful cloud adoption, which depends on its current business practices in each theme. These business practices can be in one of three possible phases:

- Tactical – This is where workloads are currently operating individually but not cohesively with a strategy for future growth. At this phase, a company is focused on keeping costs of individual systems as low as possible and moving onto a cloud system with the least possible disruption of operations. At this phase there is no provisioning done for future scalability.
- Strategic – This is when a wider strategy is used to govern individual workloads with an intent to scale to future needs. This is when a company has started accepting the changes necessary for cloud adoption and has involved its processes and people. At this phase, the company's IT department has efficiency and effectiveness which ensure greater value for a company through its cloud adoption for business.
- Transformational – This is where a company has its cloud operations running correctly and there is a focus on incorporating insights and data gathered from using the cloud. At this phase, a company undergoes transparent sharing of data, gathering and analysis of new data, use of machine learning (ML) for predictive and prescriptive analytical functions, and a transformation of both people and processes. This fosters further changes to technology. Additionally, the IT systems for a company go from being a cost center to a business partner and an engine for innovation. This phase also features blameless analysis at the C-level and development teams that are autonomous.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #24 of 60

Question ID: 1495902

Your company needs to assess its readiness for adopting the Google Cloud platform. For this, you need to assess its cloud readiness. Which theme of the Google Cloud Adoption Framework covers the ability of a company to leverage services that are native to the cloud to reduce overhead in its IT operations?

- A) Learn

✖
- B) Lead

✖
- C) Scale

✔
- D) Secure

✖

Explanation

The correct option is scale. This is the theme that covers the level to which a company uses cloud services, provisions workloads, performs automation, and maintains the applications it needs for its business. This theme covers how a business uses native cloud technologies, like serverless computing and automation, to improve its operations and scalability.

The other options are incorrect as they do not measure a company's depth of involvement in leveraging cloud-native systems to improve its operations. The other options are also a part of the themes of the Google Cloud Adoption Framework (CAF) and address different aspects of the cloud readiness of a company. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technology that are involved in a company planning a cloud adoption. There are three components to Google's CAF:

- Three maturity phases
- Four cloud adoption themes
- Seventeen epics

A component of the CAF is the four themes:

- Learn – This measures the learning systems in place for an organization to upskill its IT workforce. It also includes the ability of a company to leverage partners who are skilled in cloud technology to augment its existing workforce.
- Lead – This measures the level of support that the IT workforce receives from the company's leadership for cloud migration. It also covers the level of motivation, collaboration, and cross-functionality of the teams. The team structures, executive sponsorship, and budgeting for cloud projects are also covered in this theme.
- Scale – This measures the depth of usage of services that are native to the cloud for reducing overhead in operations, as well as the level of automation you perform for policies and processes which were operated manually. This theme covers how cloud services and resources are provisioned, capacities of workloads distributed, and updates for applications performed.
- Secure – This measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric. This covers the access control systems in place as well as the security strategies and technologies that you use to protect your services.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #25 of 60

Question ID: 1497140

You need to create a system that can provide intuitive and high quality conversations for customers through voicebots and chatbots. What Google technology would you select for this need?

- A) Visual Inspection API

✖
- B) Document AI

✖
- C) Media Translation API

✖
- D) Dialogflow

✔

Explanation

You would use Dialogflow in this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It can also be used for digital platforms, including Slack, Facebook Messenger, and Google. The Dialogflow Customer Experience Edition (CX) enables the advanced development of chatbots and voicebots. It contains a visual platform for building bots, tools for modularizing bots, support for advanced interactive voice response (IVR), as well as tools for versioning and collaboration. It has been optimized and scaled for use at an enterprise-level.

Document Artificial Intelligence (AI) is not the appropriate choice for this scenario. You would use Document AI for the classification and extraction of structured data from existing documents. This can enable you to find key insights and automate organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gRPC) framework.

You would not use Visual Inspection API for this scenario. You would use Visual Inspection API for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables development of advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Dialogflow](#)

Related Material

Question #26 of 60

Question ID: 1495931

A company uses custom made software to provide business services for consumers. The company is having trouble related to revealing certain aspects of its third-party applications to customers. How can it prevent the leakage of these vendor-related interfaces to its customers?

- A) Use a monolithic self-made application with no APIs

✖
- B) Block all exposed APIs to ensure confidentiality

✖
- C) Allow customers to install vendor-specific apps

✖
- D) Use API endpoints and move the code to the cloud

✔

Explanation

The company needs to use application programming interface (API) endpoints and move the code to the cloud. By utilizing endpoints that are vendor-agnostic and moving the entire application to Google Cloud Platform (GCP), you can ensure that any interfaces that are specific to vendors and third parties are not revealed to customers. You can move the application code to GCP using Migrate for Anthos.

The company should not allow customers to install vendor specific apps. You need to maintain a level of abstraction between customers and the vendors that create the applications you use to provide services to customers. You need to ensure that the customers do not have direct access to these applications and instead only reveal the necessary services to them using APIs.

The company should not use a monolithic self-made application with no APIs. APIs help move away from a monolithic architecture to application development and move to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

The company should not block all exposed APIs to ensure confidentiality. Consumers need access to APIs to perform various business tasks, so blocking access to the APIs would disrupt business operations. Instead, you need to provide a more controlled solution by first creating APIs that are vendor-agnostic. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #27 of 60

Question ID: 1495908

You work as a cloud consultant for a manufacturing firm that wants to use cloud-based business intelligence (BI) tools to identify issues in its production lines. Which of these would you need to do to ensure that your company makes the best use of its BI tools?

- A) Opt for an on-premises BI tool for future scalability

✖
- B) Normalize and clean the data for BI

✔
- C) Keep data for BI siloed in different systems

✖
- D) Encourage all users to use open source BI tools

✖

Explanation

You would need to normalize and clean the data for business intelligence (BI). For data to be useful for BI, you need to ensure it is high quality, updated, and cleaned. BI is a process of using technology to collect and study organizational data for making strategic organizational decisions. This involves collecting the data into a storage repository like a data warehouse and using tools to perform data analysis on the stored data. Looker is a BI tool offered by Google Cloud. Some examples of BI applications include analysis of the shopping preferences of customers, costs of company operations, and sales data. BI is also useful for comparing a company's operations versus pre-established standards and baselines. There are three categories of BI tools:

- On-premises – These are used inside an on-premises data center with data warehouses also on-premises. They lack the scalability available in cloud-based BI tools.
- Open source – These are affordable tools and can help save on the costs of infrastructure when used as cloud-based variants. However, they need considerable expertise and coding skills to be used.
- Cloud-based – These are highly effective tools for working with data in high volumes and streaming formats. They are also economical as the technical expertise and infrastructural overhead required for their use is the responsibility of the cloud service provider (CSP).

You would not keep data for BI siloed in different systems as this does not allow users access to all of the data required for BI purposes. Siloed data is distributed across multiple systems and cannot be accessed by other systems due to outdated software or stringent user access permissions. This makes key data needed for BI become inaccessible. Using an Extract, Transform, and Load (ETL) tool can help consolidate the data from across multiple systems needed for BI applications.

You would not encourage all users to use open source BI tools. You need to use managed tools for BI as much as possible because it reduces the complexity of using the tools and ensures that a lack of programming expertise or technical know-how does not become a performance bottleneck in the BI process. Open source BI tools require technical knowledge and may not be suitable for all users inside an organization. You would instead opt for a managed cloud-based BI tool, such as Looker.

You would not opt for an on-premises BI tool for future scalability because on-premises systems are not as scalable as cloud-based solutions. For accommodating future growth and working with massive datasets, you need a cloud-based solution that allows you access to managed data warehouses and BI tools with various features built in for ease of use.

You need to know of some key ways an organization can utilize BI tools, including:

- Reporting of key data for creating strategy and initiatives for the business
- Mining data through massive datasets for identifying trends and patterns
- Providing visualization of data for easy comprehension of complex data
- Analyzing real-time and streaming data, which may be gathered from sources, such as Internet of Things (IoT) sensors and stock market data
- Using predictive modeling and analysis for forecasting future trends using historical data
- Measuring business performance by analyzing sales data and customer feedback

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify common Google Cloud solutions for smart analytics

References:

[Cloud.google.com > What is Business Intelligence?](#)

Related Material

Question #28 of 60

Question ID: 1497127

You need to recommend a Google product to a company that wants to enhance its applications by incorporating intelligent components that are plug-and-play. For this, the company needs to create, deploy, and scale its machine learning models quickly using a single platform. Which Google product should you recommend for this scenario?

- A) Cloud Life Sciences**
- B) Memorystore**
- C) Network Intelligence Center**
- D) Vertex AI**
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Explanation

You should recommend Vertex Artificial Intelligence (AI) for this scenario. Vertex AI provides a singular application programming interface (API) platform for building and scaling machine learning (ML) models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. Vertex AI has various services and tools within it, such as Automated Machine Learning (AutoML), which is a system that allows you to perform training for ML models without the need for coding. Vertex AI also allows you to create ML models that are advanced using its custom model tooling. It has Machine Learning Operations (MLOps) tools that enable ML model maintenance using the Vertex AI Pipelines, which streamlines the process of running pipelines for ML, and the Vertex Feature Store, which you can use to serve and use AI technologies as features for your ML applications. ML is the science of developing intelligent systems and applications that can “learn” from existing data provided to it and it can then provide services that are intuitive, dynamic, and adaptive. ML is a subset of AI that involves methods for making a machine mimic the ability of a human to make decisions and solve problems. This is done using deep learning and neural networks that enable the ML system to learn continuously. It removes the need to program the ML system explicitly by providing it with large quantities of data.

You would not recommend Network Intelligence Center for this scenario. Network Intelligence Center is used to monitor, verify, and optimize a network. It can be used for Google Cloud, on-premises, and multi-cloud systems. Network Intelligence Center reduces the risk of outages occurring and ensures greater levels of compliance with regulations and security.

You would not recommend Memorystore for this scenario. Memorystore is a fully managed service that provides in-memory storage of data with less than a millisecond of response time to access the data. It does this by deploying distributed caches for applications. Memorystore helps migrate the cache layer to the cloud without needing to modify existing application code.

You would not recommend Cloud Life Sciences for this scenario. Cloud Life Sciences enables you to manage, process, and transform data related to life sciences. This data can include genomics and biomedical data. Cloud Life Sciences provides workflows that are containerized and allow for the processing, analysis, and annotations of life sciences data with scalability.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify Google Cloud's solutions for machine learning and AI

References:

[Cloud.google.com > Vertex AI](#)

Related Material

Question #29 of 60

Question ID: 1495243

Your company runs several workloads using containerized applications with specific OS and networking configurations. You need to create and manage clusters with granular control at the nodal level and have a way of automatically running the clusters without monitoring.

How should you do this?

- A) Cloud Functions**
- B) GKE**
- C) Compute Engine**
- D) App Engine**
- 
- 
- 
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Explanation

You should use Google Kubernetes Engine (GKE) in this scenario. GKE provides a fully managed Kubernetes platform that has scalability and automation. It provides granular control over containers so you can specify the exact OS, memory, storage disk, CPU, and networking configuration you need. GKE has a fully managed Autopilot mode that manages the complete infrastructure of a cluster without the need for monitoring or configuration. This mode also allows for cost optimization by ensuring billing is done only for utilized pods and not for system overhead, components, or unused resources. The Standard mode of GKE allows you to implement more granular controls and have customized workloads for administrative tasks. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Google Kubernetes Engine](#)

Related Material

Question #30 of 60

Question ID: 1495910

For your company's data-driven operations, you are looking for a way to organize all the organizational data in the company by various business domains, which include customer support, human resources, and sales, and use a decentralized structure in a way that provides federated data ownership. Which of these options would be the most appropriate for this requirement?

- A) Data lake

✖
- B) Data lakehouse

✖
- C) Data mesh

✔
- D) Data warehouse

✖

Explanation

A data mesh would be the most appropriate choice for this requirement because a data mesh provides a decentralized system architecture for organizing data by business domains. This ensures greater ownership for the people that produce each dataset. Producers can then specify policies for data governance. These policies enable greater self-service by users across the organization through the emphasis on clear documentation, quality concerns, and access systems. Using a data mesh requires cloud adoption for the necessary scalability and achievement of data management goals. Using a data mesh removes operational hurdles associated with more centralized and monolithic data systems. A data mesh provides federated ownership of data and allows data access by multiple teams through the use of application programming interfaces (APIs).

You would not use a data lakehouse in this scenario. A data lakehouse is a data platform that combines data lakes and data warehouse technologies in a way that uses the tools and management features of data warehouses and merges them with the flexibility of storing unstructured data in data lakes. You take away the overhead that is involved with using data warehouses and data lakes by using a data lakehouse. A data lakehouse provides a single point of access to both real-time and batch data, and also provides ways to implement data governance.

You would not use a data lake in this scenario. A data lake allows you to store structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstreams, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources, including edge computing, cloud, or on-premises systems. The data can be stored in its original format and processed as a batch or in real-time. This data can then be analyzed using tools like Python, SQL, and R.

You would not use a data warehouse in this scenario. A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities like complex analytics, which can lead to making informed management decisions. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Emerging Trends](#)

Related Material

Question #31 of 60

Question ID: 1495287

A company has an SLA in place for its service and wants to ensure that it is met. What can be the potential impact of unexpected downtime on the system? (Choose two.)

- A) Greater error budget

✖
- B) Financial losses

✔
- C) Addition of new features

✖
- D) Increased ROI

✖
- E) SLA not being met

✔

Explanation

System downtime can lead to service level agreements (SLAs) not being met, which can then incur financial losses for a company. SLAs are an agreement between the service provider and customers that specifies the level of availability that the service will have. To ensure that an SLA is met, you need to keep a service level objective (SLO) value that is higher than the SLA. This helps you get early warnings when there is a danger of not meeting the SLA. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

The addition of new features is not a potential impact of downtime. During system downtime, there cannot be new features added. Developers may want to add new features to the system during application development, while operations teams want to minimize any possible disruptions to the service and seek to minimize any modifications to the service. This creates a conflict between the teams. Creating a common SLO ensures that there can be an agreement between operations and development teams to meet a common goal of ensuring service uptime while adding new features.

An increased return on investment (ROI) is not a potential impact of downtime. The ROI for a company can be adversely affected by prolonged or unexpected downtimes, as this can cause customers to leave. Monitoring costs and measuring ROI are key functions of business teams and management and are important to ensure that the organization's financial performance is in line with the strategy and cost management principles of the company.

A greater error budget is not a potential impact of downtime. The occurrence of prolonged system downtime indicates that the error budget is exhausted. The error budget is the measure of the system's possible downtime, which is the difference between the SLO and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #32 of 60

Question ID: 1495176

A company currently organizes training for upskilling its employees and relies on the assistance of third-party companies to run its training sessions. How can the company move from this phase to a transformational cloud adoption maturity phase?

- A) Encourage self-teaching for employees and bring in more third parties for training

B) Reduce peer-based knowledge sharing and bring in third-party support for learning

C) Reduce training within the company and increase training from third parties

D) Reduce third-party involvement to a minimal level and increase knowledge sharing through peers
- ✖

✖

✖

✔

Explanation

The correct approach to move the company to a transformational phase in the Cloud Maturity Scale is to reduce third-party involvement to a minimal level and increase knowledge sharing through peers. A key aspect of the transformational phase in the Cloud Maturity Scale is increasing knowledge sharing and lateral awareness while ensuring individuals and teams can make more autonomous decisions.

The other options are incorrect as a key aspect of the transformational phase is to reduce the involvement of third parties to only augment what is done internally in the company through raised awareness and peer-based knowledge sharing.

A company's Cloud Maturity Scale is formed when the four themes are evaluated using the three phases:

- Tactical
 - Learn – Learning is mainly self-taught, and there is reliance on third parties for knowledge.
 - Lead – Each function's team has a leader, and there is a strong project manager.
 - Scale – Changes are risky, slow, and have high operational overhead.
 - Secure – The public Internet is avoided for security reasons, and the private network is preferred.
- Strategic
 - Learn – Learning is organized through training and can have assistance from third parties.
 - Lead – There are cloud teams that operate across functions.
 - Scale – Scaling is templatized, which maintains proper governance without the need for manual supervision.
 - Secure – The network is a hybrid one, and there is a centralized system for identity management.
- Transformational
 - Learn – Learning occurs through peers and knowledge sharing. Third parties only augment the ongoing learning culture.
 - Lead – There is more autonomy for teams and there are feature teams that operate across functions.
 - Scale – Changes are low risk, constant, and can be easily remediated, if required.
 - Secure – This is a low trust model where only the appropriate devices, users, and services are trusted.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #33 of 60

Question ID: 1495229

You need to find the right cloud infrastructure option for migrating from a legacy system with specific licensing needs and running your workloads. Which Google service should you use when you need to have granular control over infrastructural resources, such as CPU and RAM, on a particular OS?

- A) Compute Engine

B) Cloud Run

C) Cloud Functions

D) App Engine
- ✔

✖

✖

✖

Explanation

You should use Compute Engine for this scenario. You would use Compute Engine when migrating from a legacy system that has specific OS and licensing needs. Compute Engine provides the use of virtual machines (VMs) where you can specify the exact memory, CPU, disk storage, and graphics processing units (GPUs) you need, along with the exact OS and other applications you need to use.

You would not opt for App Engine in this scenario. App Engine provides a serverless platform that is fully managed for building and hosting full web applications. You can write a web application using a preferred language and then use App Engine to manage the tasks related to application and database scaling as well as performing updates to the application. App Engine is the preferred option for when you need to focus on coding a web application or a backend for a mobile app and leave the tasks of deployment and scaling to Google Cloud.

You would not opt for Cloud Functions in this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not opt for Cloud Run in this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for HTTP, Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. Another related container technology is Google Kubernetes Engine (GKE), which allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Where should I run my stuff? Choosing a Google Cloud compute option](#)

Related Material

Question #34 of 60

Question ID: 1495242

Your company uses a hybrid cloud architecture with several of its applications needing to run on its on-premises data center as well as on Google Cloud. You need to recommend a service that can meet this requirement. Which service should you recommend?

- A) App Engine

B) Firestore

C) Compute Engine

D) Anthos
- ✖

✖

✖

✔

Explanation

You should recommend Anthos. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), create and deploy apps using Google Kubernetes Engine (GKE), VMs, as well as landing zones that are serverless. Anthos allows you to use its platform to deploy both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application’s lifecycle, Anthos provides protection for both a software supply chain and an application.

You would not recommend App Engine. This cannot be used on-premises because it is a cloud computing platform. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning.

You would not recommend Compute Engine. This cannot be used on-premise because it is Infrastructure as a Service (IaaS). Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google’s technological infrastructure.

You would not recommend Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database which is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #35 of 60

Question ID: 1495926

Which Google Cloud solution can help your company modernize its infrastructure by removing downtime, reducing RTO, and preventing data corruption and loss?

- A) BigQuery

B) Cloud Build

C) Cloud Backup and DR

D) Cloud Run
- ✖

✖

✔

✖

Explanation

You should use Cloud Backup and Disaster Recovery (DR) to meet these requirements. This service provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups. This is enabled by the ability to instantly mount and use VMware engines and databases from Cloud Backup and DR backups. The incremental-forever system used by Cloud Backup and DR ensures that duplicate data does not need to be stored and results in reduced cost of storage and faster recovery times. This is made possible through changed-block tracking, which reduces storage requirements and speeds up recovery.

You would not use Cloud Run for this scenario. You would use Cloud Run when you need to perform a deployment of containerized applications, which are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling.

You would not use Cloud Build for this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD, which prevents the deployment of any images that are unsafe. Some key features of Cloud Build include:

- Fast build time using VMs with fast CPUs provisioned to cache source code and dependencies.
- Deployment automation using pipelines with integrated use of Firebase, Google Kubernetes Engine (GKE), Cloud Functions, and App Engine.
- Multicloud deployment using multilanguage support for builder images used and containerized tasks, which are portable across clouds.
- Strong privacy through Google Cloud security, which allows the use of repositories for private source code inside private networks.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Backup and DR Service](#)

Related Material

Question #36 of 60

Question ID: 1497157

Your company's accounting department has raised an issue regarding the high operating costs of the cloud deployment. You are asked to find ways to optimize the costs of running the cloud. What should you do to reduce the Total Cost of Ownership (TCO) using Google Cloud?

- A) Utilize historical usage data for monitoring

✖
- B) Use larger VM sizes than needed

✖
- C) Delete idle VM instances

✔
- D) Store data on the most available storage classes

✖

Explanation

You should reduce the Total Cost of Ownership (TCO) through cost optimization tactics, such as deleting idle virtual machine (VM) instances. Cost optimization involves using managed services, keeping a consumption model, and analyzing expenditure and attributing it correctly. Cloud monitoring dashboards display key system metrics graphically in real-time, offering an updated view of the cloud system. This helps you get a clear analysis of how your systems infrastructure, applications, and services are doing. This way, you can use metrics and logs to track and resolve issues swiftly, thus improving your TCO and reducing Mean Time to Recovery (MTTR). You can also create alarms based on metrics. An example is ensuring that server sizes are selected to match actual computing needs in order to ensure greater cost-effectiveness.

You would not reduce the TCO by using larger VM sizes than needed. Using larger server sizes than required will cause a waste of resources and thus a waste of capital. Instead, you can opt for rightsizing VM instances based on your specific requirements. For this, you can create a custom machine type that allows you to specify the exact CPU and RAM configuration for your specific processing needs.

You would not reduce the TCO by using historical usage data for monitoring. Historical data cannot give you an updated view of a cloud system. Monitoring involves observing important system metrics in real-time using dashboards. This allows you to find ways to optimize costs through more efficient use of cloud resources. An example is monitoring VM instances to identify resources that may be idle.

You would not reduce the TCO by storing data on the most available storage classes. Using the most available storage class is suitable for high availability applications. However, it is also the most expensive, and not all types of data require high availability. By using more suitable storage classes for specific types of data, you can optimize storage costs and save on TCO. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #37 of 60

Question ID: 1497159

You work for a multinational company that has recently performed a cloud adoption process. The company needs you to find ways to have more transparency into cloud expenses for better management. What cost management strategy should you use in this scenario?

- A) Increase scrutiny of on-premises technology and services

✖
- B) Delegate responsibility of cloud budgeting to one stakeholder

✖
- C) Facilitate collaboration between financial, IT, and marketing teams

✔
- D) Identify any cloud cost overruns to the company's error budget

✖

Explanation

You should facilitate collaboration between financial, IT, and marketing teams. This is necessary to ensure visibility and transparency as part of cost management. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is necessary for efficient cost management. Reducing budget overruns and unauthorized resource use can be done with more collaboration between IT and finance teams.

You would not delegate the responsibility of cloud budgeting to one stakeholder. Keeping a single person in charge of managing cloud costs cannot work for a large organization due the scale and complexity involved. An individual can look after cloud cost management only for very small companies. Large companies will require cross-team collaboration for effective cost management.

You would not identify any cloud cost overruns to the company’s error budget. An error budget is not related to cloud expenses. It is the variation between the service level agreement (SLA) and service level objective (SLO). The error budget is the measure of the system’s possible downtime, which is the difference between the SLO and the total availability figure of 100%. The SLO is kept at a higher value than the SLA to ensure early warnings of possible availability issues.

You would not increase scrutiny of on-premises technology and services. The technology and services operating inside on-premises systems do not have an effect on the costs of running a cloud deployment. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #38 of 60

Question ID: 1497150

Your company has various teams and departments, all of which concurrently use resources on a Google Cloud deployment. How should you implement financial governance on Google Cloud to allow your company to achieve more accountability in its cloud operations?

- A) Managing inventory for on-premises systems**
- B) Ensuring all resources are owned by a single user account**
- C) Using cloud resource labels and hierarchies**
- D) Keeping a flat hierarchy with a single organizational level**

Explanation

To achieve greater accountability in your cloud deployment for financial governance, you should use cloud resource labels and hierarchies. Using resource hierarchies simplifies the allocation of costs to specific departments and ensures accountability. This action also provides insights into the return on investment (ROI) for your cloud deployment. A resource hierarchy enables you to create a structure for your Cloud account based on your specific business needs. The hierarchy can be described below:

- Level 1 (topmost) – Organization: The Company
- Level 2 – Folders: Departments, shared infrastructure, teams within departments, and products within teams
- Level 3 – Projects: Development, testing, and production using Google Cloud Platform (GCP) for projects
- Level 4 – Resources: Compute Engine instances, Cloud Storage buckets, and App Engine services

Additionally, resource labels can be used to group several resources together and to identify the teams that own particular resources. As an example, resource labels can be App:UI, Env:Prod, and Dept:Testing, which indicate that a resource is for an app called UI in the production environment used by the testing department.

A company needs various financial governance tools for easy cost management that ensures clarity, accountability, and control. Google Cloud financial governance helps a company control costs by providing various tools and services.

- Clarity – This is achieved using billing reports from the Google Cloud Platform (GCP) console, billing application programming interfaces (APIs), and intelligent recommendations.
- Accountability – This is achieved using resource hierarchies and resource labels.
- Control – This is achieved using organizational policies and cost management controls.

Managing inventory for on-premises systems is not a way of implementing accountability on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Ensuring a single user account owns all resources is not a way of implementing accountability on a cloud deployment because a cloud deployment for a company can have several different departments using the system. For this, there needs to be a way to have transparency so the company knows the exact users and departments that utilize specific cloud resources.

Keeping a flat hierarchy with a single organizational level is not a way of implementing accountability on cloud deployment. Keeping a multi-level hierarchy that reflects the operating structure of an organization is critical for providing accurate financial accountability in the cloud deployment. By having various levels of scale, costs can be allocated correctly and managed efficiently based on the specific departments and projects that utilize the resources. With a flat organizational hierarchy, it would be impossible to have accountability for resource utilization and the associated costs of specific departments and projects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #39 of 60

Question ID: 1495929

Your company has an on-premises data center that it uses to provide data processing services to customers. Due to the nature of the legacy system, there have been some business disruptions to the current users of specific services due to network traffic issues, anomalies, and malicious bot attacks.

What would be the best way to solve this issue?

- A) Create Kubernetes clusters with GKE**
- B) Move apps to the cloud using VMware engine**
- C) Use API management with Apigee**
- D) Build a document database with Firestore**

Explanation

You should use application programming interface (API) management with Apigee to solve this issue. API management with Apigee enables you to perform monitoring and analysis of API traffic to help mitigate key problems related to performance. It also allows you to build a layered defense system that includes distributed denial of service (DDoS) attack protection, bot handling, API guarding, as well as web application firewalls (WAFs). Apigee lets you monitor APIs and ensure your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. API management with Apigee allows you to perform routing based on context, transform payloads, and perform API facades that are user specific. You can start by routing a small portion of the users to the modernized version of the service that uses APIs, and then based on the user feedback and experience, you can route all the users to the API-based modern system.

You would not create Kubernetes clusters with Google Kubernetes Engine (GKE) for this scenario. GKE is not used for API management, which is what this scenario needs. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not build a document database with Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not move apps to the cloud using VMware engine. VMware Engine can be used to move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #40 of 60

Question ID: 1495909

You are in the process of designing a data warehousing solution for a company. Which of these would be the key characteristics of the data warehouse? (Choose two.)

- A) Provides non-persistent storage

✖
- B) Allows SQL for data access and manipulation

✔
- C) Provides a way to access unknown data

✖
- D) Contains unstructured data, which is accessed using code

✖
- E) Answers known queries by accessing known data

✔

Explanation

The following options are the correct characteristics of a data warehouse:

- Answering known queries by accessing known data
- Allowing SQL for data access and manipulation

A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities, such as complex analytics, which can lead to making informed management decisions.

The following options are not characteristics of a data warehouse but of a data lake:

- Unstructured data, which is accessed using code
- Providing a way to access unknown data

A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstreams, databases, and transactional information.

A data warehouse is not used for providing non-persistent storage but is a storage repository for the organization's data. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Data warehouse or data lake?](#)

Related Material

Question #41 of 60

Question ID: 1495288

Your operations team spends a lot of time on repetitive tasks. You want to bring greater efficiency into your Google Cloud operations by reducing toil. What would be the best strategy to achieve this goal?

- A) Perform encryption of data at rest

✖
- B) Create new SLAs with users

✖
- C) Halt deployments

✖
- D) Use more observability tools

✔

Explanation

You should use more observability tools to reduce toil. Toil comprises manual and repetitive tasks, which should be automated so that team members can focus their time on higher value work. By using observability tools like Cloud Tracing, Cloud Monitoring, and Cloud Logging, you can gain more insight into services and ensure their correct automation. Creating more automation of services can help reduce toil in an organization, which boosts work efficiency and productivity. A Site Reliability Engineering (SRE) recommended best practice is for operations teams to look into issues with services and attempt to identify the reasons for failure. For this, the teams should gain more knowledge on how a service operates.

You would not perform encryption of data at rest to reduce toil. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure.

You would not create new service level agreements (SLAs) with users to reduce toil. SLAs are an agreement between the service provider and customer that specifies the level of availability that the service will have. Reduction of toil involves creating more automation of systems, which removes the need for manual involvement of team members.

You would not halt deployments to reduce toil. Halting deployment is not done for toil reduction but is done in situations where an error budget is consumed. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. Once an organization reaches a stage where the error budget for a given period is exhausted, then management will need to halt the development of new features until the error budget is restored.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #42 of 60

Question ID: 1495166

Your company is looking at ways to change its operation and resource utilization so that it excels at delivering the maximum possible value to customers. For this, it is looking into performing digital transformation using cloud technology.

Which of these is a key part of the journey to digital transformation?

- A) Segregating customer feedback from product development activities

✖
- B) Guiding business decisions using insight from data analytics tools

✔
- C) Keeping team collaboration focused within each company department

✖
- D) Using legacy on-premises systems for quickly deploying new global applications

✖

Explanation

Guiding business decisions using insight from data analytics tools is a key part of the journey to digital transformation. Experimentation with new technologies and concepts can help a company gain more insight into market needs and customer preferences. This experimentation can then be analyzed using advanced data analytics tools to make the right business decisions for growth.

Segregating customer feedback from product development activities is not a key part of the journey to digital transformation. This is because customer feedback needs to be incorporated into product development efforts and not segregated from them so the company can create offerings that better serve customers. Focusing on customers and their preferences through analysis of customer usage data can help create products that provide more value to customers and lead to greater business growth.

Keeping team collaboration focused within each company department is not a key part of the journey to digital transformation. Digital transformation involves using tools to share knowledge and collaborate across a company's organizational boundaries. Team collaboration and knowledge sharing eventually leads to more autonomy in decision making and creates a more dynamic workforce.

Using legacy on-premises systems for quickly deploying new global applications is not a key part of the journey to digital transformation. Digital transformation involves using modern technology that has the flexibility to enable you to deploy applications onto the cloud swiftly and manage them efficiently.

Digital transformation involves utilizing new technology to change the design and definition of a company's relationships with its partners, customers, and employees. Digital transformation involves using new digital systems including private, public, and hybrid clouds to change existing culture, work processes, and customer experiences with an aim to meet the changing dynamics in businesses and markets. Some key benefits of digital transformation include:

- Modernized infrastructure – This includes using serverless systems, containers, and global cloud networks for an efficient and effective IT infrastructure.
- Better data management – This include using new systems and tools to manage massive amounts of data which originate from various devices and sources.
- Stronger insights – This covers using advanced business analytics tools to provide useful data insights.
- Faster collaboration – This involves dismantling the traditional team silo concept where a team is cut off from other teams and instead using new tools that allow teams to collaborate across various global locations swiftly. Faster collaboration helps create value and produce results for customers more quickly.
- Finding business solutions – This involves using new systems, processes, and technologies for solving business problems and identifying ways to improve how businesses interact and work with their customers.
- Savings on costs – This involves using tools for making processes more efficient and identifying the high and low performance aspects of a business accurately.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is Digital Transformation > What are the benefits of digital transformation?](#)

Related Material

Question #43 of 60

Question ID: 1497149

A bank has an on-premises data center that it needs to maintain to ensure regulatory compliance. It also has several applications that it uses for internal purposes as well as for serving its customers. What should you use to provide a single portal for visibility of their nodes and clusters in their cloud and on-premises workloads that also manages large volumes of Kubernetes clusters inside their applications?

- A) App Engine

✖
- B) Firestore

✖
- C) Anthos

✔
- D) Cloud Functions

✖

Explanation

You should use Anthos for this scenario. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), and create and deploy apps using Google Kubernetes Engine (GKE), VMs, and landing zones that are serverless. Anthos allows you to use its platform for deploying both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application’s lifecycle, Anthos provides protection for both a software supply chain and application.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #44 of 60

Question ID: 1495183

Your company is creating a new e-commerce portal and wants to begin the process of data classification and management for it. You need to identify all the possible structured and unstructured data you will have to work with. Which of these is an example of structured data?

- A) Reviews written for products by customers**
- B) Customer ratings for each product**
- C) Pictures of products featured on the site**
- D) Descriptions of the products written by vendors**

Explanation

Customer ratings for each product are an example of structured data. This is because a product rating can be a numerical value or score. Credit card numbers and product names are also examples of structured data. Structured data is data that is organized and quantitative. This allows for easy storage and management using a database. Databases provide an organized way of storing data as tables that can be accessed using a computing system, and can ingest large amounts of data from various sources. The consistency and accuracy levels of data stored in a database comprise the transaction integrity.

The other options are not examples of structured data. They are unstructured data. Product reviews and descriptions both comprise text files, and product pictures are image files. Image files and documents from word processing applications are examples of unstructured data. Unstructured data comprises data that is qualitative and is stored as objects. This allows the data to be stored in its original format along with key metadata that provides ways to uniquely identify the data. Metadata is data about stored objects.

Some additional terms you need to know include:

- Data point – This represents a chunk of information, which can be customer purchases.
- Dataset – These are an aggregation of several data points, which belong to a particular category like “transactions by customers”.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for data management

References:
[Services.google.com > Cloud Digital Leader Study Guide](#)

Related Material

Question #45 of 60

Question ID: 1495293

In an effort to have greater control over the performance of applications running on the Cloud system, a company’s operations team wants to investigate the workings of the applications. What Google Cloud service should your operations team use to get a better understanding of what occurs when requests traverse through the system?

- A) Cloud Armor**
- B) Cloud Trace**
- C) Cloud Logging**
- D) Cloud Monitoring**

Explanation

Your operations team should use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app. Application Performance Management (APM) is a combination of several tools that ensure optimal performance of applications. APM integrates tools to monitor and troubleshoot applications, which include Cloud Monitoring and Cloud Logging. It also includes latency and cost reduction systems like Cloud Trace and Cloud Profiler. This boosts the efficiency of operations.

You would not use Cloud Monitoring for this scenario. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications.

You would not use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets for storing and processing log data based on regional compliance requirements.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Trace](#)

Related Material

Question #46 of 60

Question ID: 1495193

A company that is in the digital publishing domain has approached you to help them find a suitable technology for building a data warehouse that they can use for performing analysis and reporting with scalability that is seamless. Which Google Cloud service should you recommend for this requirement?

- A) DataFlow

B) Cloud Storage

C) Cloud Trace

D) BigQuery
- ✖

✖

✖

✔

Explanation

You should recommend BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture allowing you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not recommend DataFlow for this scenario because it is not a data warehousing service but a service for performing data processing. DataFlow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not recommend Cloud Storage for this scenario because it is an object storage service and not a data warehousing service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not recommend Cloud Trace in this scenario as it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is BigQuery?](#)

Related Material

Question #47 of 60

Question ID: 1495220

Your company currently has a legacy on-premises data center, which it uses to provide online shopping services to its customers. Which of these features will require the company to modernize its IT infrastructure with Google Cloud?

- A) Increasing the physical security of the data center

B) Connecting legacy systems to the Internet using VPNs

C) Deploying global applications that are fault tolerant and highly available

D) Performing vertical scaling of data center infrastructure
- ✖

✖

✔

✖

Explanation

Deploying global applications that are fault tolerant and highly available requires modernizing IT infrastructure with Google Cloud. A legacy infrastructure cannot deliver a global fault tolerant infrastructure at the level and scalability of a cloud deployment. Google Cloud provides a global infrastructure that consists of regions and zones. This gives you the flexibility to choose how you want to deploy your applications for the best levels of reliability and availability. It also provides the least latency possible for end users through the use of edge locations. Regions are geographical areas that are independent. Each region consists of zones, which are areas of deployment of Google resources inside of a region. A zone can be considered as a single failure domain, which implies that for an application to have fault tolerance and high availability, it needs to be deployed across multiple zones. The services of Google's infrastructure are available globally and have round the clock maintenance and upgrades performed. Google's global infrastructure makes the process of managing multi-region deployments simple and provides flexible load balancing features.

Performing vertical scaling of data center infrastructure does not require a cloud deployment. Vertical scaling involves increasing the computing power of existing machines and can be done within a data center itself. This is not a cloud exclusive feature. However, a cloud deployment would be more appropriate if there was a requirement for computing resources to be adjusted dynamically as per changing user demand. This is a concept called elasticity, which ensures that the optimum amount of cloud resources is allocated for handling workloads.

Connecting legacy systems to the Internet using virtual private networks (VPNs) does not require a cloud deployment. For connecting computer systems to the Internet, you need an Internet connection through an Internet Service Provider (ISP), and a VPN connection through a VPN provider. However, modernizing the infrastructure with Google Cloud does have the advantage of improving the security of the existing systems through strong identity and access management as well as network protection tools to safeguard against Internet-based malware.

Increasing the physical security of the data center does not require a cloud deployment. The physical security of the data center is the responsibility of the company and needs to be done by the company itself. The security measures will include security barriers, such as walls and fencing; and monitoring systems, such as cameras and sensors. Security measures might also include guards and card- or token-based access control.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Geography and regions](#)

Related Material

Question #48 of 60

Question ID: 1495224

You need to help your company choose an appropriate cloud architecture for its operations. Which of these is an advantage of adopting a hybrid cloud architecture?

- A)** Ending data center costs by migrating infrastructure to a public cloud
- B)** Availing services from multiple public cloud service providers
- C)** Maximizing security of sensitive data through on-premises storage
- D) Safeguarding all applications and data from public clouds**

Explanation

Maximizing security of sensitive data through on-premises storage is an advantage of adopting a hybrid cloud architecture. By using a hybrid cloud architecture, you can provision resources from a cloud service provider (CSP) and not have to procure the same computing resources for the data center. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. Some key advantages of adopting a hybrid cloud architecture include greater agility in handling usage bursts, leveraging new and advanced analytics technology, and speeding up time-to-market and reducing cycle times for new applications.

Availing services from multiple public CSPs is not a feature of hybrid cloud architectures but of multi-cloud architectures. A multi-cloud deployment involves using multiple public CSPs where each public cloud service has a different function for the company. This allows a company to use one public cloud for database services and another one for analytics. A company can then use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Safeguarding all applications and data from public clouds is not a feature of hybrid cloud architectures but of private cloud architectures. In this, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This system does not leverage systems that are a part of a public cloud service offering.

Ending data center costs by migrating infrastructure to a public cloud is not an aspect of hybrid cloud architectures but of public cloud architectures. This infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This infrastructure option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #49 of 60

Question ID: 1495227

You want to create a cloud-based testing environment that duplicates various different hardware setups so that the testing team can perform various application tests within this isolated environment. Which of these technologies should you use for this requirement?

- A)** Containers
- B)** Physical computers
- C) Virtual machines**
- D)** Serverless data warehouses

Explanation

This is a use case for virtual machines (VMs). A VM is a virtual computer that runs operating systems and applications. Multiple VMs can run on a single physical computer and access various resources from a hypervisor. A hypervisor is software that creates an abstraction of all the computing resources of the physical computer and provides these resources to VMs running on the computer as required. The VM architecture provides the underlying framework in use for modern cloud computing. Compute Engine is Google's VM service, which offers VMs that can be customized. Some uses of VMs in enterprises include:

- Server consolidation – VMs can act as servers to host other VMs and so reduce sprawl by consolidating multiple computing systems into a single physical machine.
- Migration of workloads – VMs are flexible and portable, which enable swift workload migration for companies.
- Test environment creation – VMs can provide an isolated system that can be used for testing without affecting the surrounding systems.
- Disaster recovery – You can duplicate an existing system as a VM on the cloud, thus adding an additional layer of security to the system.
- DevOps – VMs can be activated, deactivated, and migrated smoothly, which facilitates flexible app development.
- Hybrid systems – VMs help you flexibly integrate existing on-premises legacy systems with a cloud environment to create a hybrid computing environment.

You would not use containers for this scenario. Containers are better suited for creating applications that need to be compartmentalized and made portable across multiple platforms. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You would not use serverless data warehouses for this scenario. A serverless system like BigQuery would be suitable for a requirement where there needs to be very high levels of scalability for working with very large quantities of data. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use physical computers for this scenario. This is because procuring physical computers for creating testing environments is not as economical, feasible, or flexible as using a VM based system. A single physical computer has one hardware configuration with a specific CPU, RAM, and storage setup. Using several different VMs, you can have several hardware setups, which makes the process of testing an application across multiple platforms much simpler.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is a Virtual Machine?](#)

Related Material

Question #50 of 60

Question ID: 1497152

Your company has a large catalog with many stock keeping units (SKUs). The company wants to utilize efficient third-party tools for managing costs for projects programmatically in order to have greater clarity into cloud costs and their management. Which of these options will help meet this requirement?

- A) Billing the on-premises data center**
- B) Using billing APIs**
- C) Billing the cloud service provider**
- D) Setting resource quotas**

Explanation
Using billing application programming interfaces (APIs) will help meet this requirement. With billing APIs, you can use various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. Other methods of ensuring clarity for financial governance include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized. Also, you can use billing reports from the Google Cloud Platform (GCP) console to plan, track, and ensure optimal use of resources within your cloud deployment. This is done through views of trends for current costs and forecasts for future spending. You can also export this data to Data Studio and BigQuery to create customized dashboards that provide more details on cloud utilization and its associated costs.

You would not bill the on-premises data center in this scenario. This is not a way to provide programmatic management of project billing for a cloud deployment. For programmatic management of costs and billing, you need to utilize APIs, which can provide an interface between your systems and the services offered by third parties. The on-premises data center’s operations and cost management are unrelated to the cloud deployment.

You would not set resource quotas in this scenario. Setting resource quotas does not provide a way for programmatically managing billing for projects. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets.

You will not bill the cloud service provider (CSP). You, as a customer, are billed by the CSP based on the resource consumption and service utilization of your cloud deployment. CSPs use the Pay-As-You-Go (PAYG) model of payment, where a cloud consumer only needs to pay for resources consumed and not for unused resources. For example, if a computing instance is in a terminated state, then it will not contribute to the total billable costs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #51 of 60

Question ID: 1497138

You work for an online shopping company that allows customers to write product reviews as text on the shopping website and app. You want to enable your developers to extract a user’s sentiment from blocks of text from the user’s review to better understand customer experiences. Which of these technologies would you use for that purpose?

- A) Cloud Translation**
- B) Cloud Trace**
- C) Cloud Natural Language API**
- D) Cloud Vision**

Explanation

You would use Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language Artificial Intelligence (AI) enables you to gain understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be used by developers through API calls or through reference to a document that may be stored on Cloud Storage.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language and vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Cloud Vision is not the appropriate choice for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

Cloud Trace is not the appropriate choice for this scenario. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify Google Cloud's solutions for machine learning and AI

References:

[Cloud.google.com > Natural Language AI](#)

Related Material

Question #52 of 60

Question ID: 1495221

A game development company has been developing games from a legacy on-premises data center and is now considering making a digital transformation. How would the company benefit from modernizing its infrastructure with cloud technology?

- A)** Delegating application security to the cloud service provider
- B)** Ensuring all resources are paid for regardless of usage
- C) Massive scalability by using serverless technology**
- D)** Making significant investments in new emergent technologies

Explanation

Using cloud-based technologies like serverless computing, a company can achieve massive levels of scalability, which may not be feasible using traditional legacy systems. For example, BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Delegating application security to the cloud service provider (CSP) is not a benefit of using cloud technology. Security of applications and all the data stored in the cloud is the responsibility of the customer and not the CSP. The CSP is responsible for the security of its data centers and of the cloud.

Paying for all resources, regardless of usage, is not a benefit of using cloud technology. Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth, and allows a company to have its staff work on more strategic tasks that help grow the business.

Making significant investments in new emergent technologies is not a benefit of using cloud technology. CSPs have access to the latest technologies, which customers can avail of as services. This removes the risk of the company having to make large investments into new technologies, which may become obsolete in time.

Some distinct benefits of cloud technology are outlined below:

- Security – Through a shared responsibility model and best practices, you can ensure greater safety of your operations using Google Cloud.
- Reliability – This helps you ensure your workloads can perform correctly, handle failures, and automatically fix any issues that may arise. This also covers data backups, fault isolation, and disaster scenario preparation.
- High Availability – This ensures that your applications are always ready to serve customers swiftly with almost no downtime. A measure of availability is five 9s, which corresponds to a system being available 99.999% of the time.
- Elasticity – This is the ability to scale your computing resources to match consumer demand without having to perform resource provisioning manually or guess future capacity needs.
- Agility – This is the ability to swiftly deploy new applications by removing unnecessary operational overhead so that the development team can focus on creating and innovating products.
- Pay-as-you-go (PAYG) pricing – This is a flexible payment option where a cloud consumer only pays for resources consumed.
- Scalability – This is using the right technology to easily make your application meet growing demands on a massive scale using serverless technologies.
- Global Reach – Using edge locations and data centers around the world, Google Cloud's global infrastructure can improve user experience with low latency for end users, regardless of their location on the globe.
- Economy of scale – This is having lower PAYG pricing due to accumulated cloud usage from a very large number of cloud customers.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Learn what modernizing IT infrastructure with Google Cloud means

References:

[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Related Material

Question #53 of 60

Question ID: 1495277

To ensure the security of your cloud deployment, senior management has decided to implement the principle of least privilege for specifying user access permissions across the cloud. What will be the impact of using a resource access model based on the principle of least privilege for the security of the cloud?

- A)** Only senior management can access resources on the cloud
- B) A user only has access to the cloud resources necessary for task completion**
- C) Cloud resource access needs to be approved by the cloud service provider**
- D)** Users need authorization to use on-premises applications

Explanation

The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that users cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

The least privilege principle does not imply that only senior management can access resources on the cloud. An access management system that only allows senior management access to cloud resources would not allow for normal use of a cloud deployment and would not provide the company with the growth it seeks from using the cloud. By the principle of least privilege, all employees, regardless of their designation, can have access that is limited by their specific role within the organization.

The least privilege principle does not imply that users need authorization to use on-premises applications. The access in systems in place for an on-premises data center is not related to the Identity and Access Management (IAM) that are used on a cloud deployment for restricting access. The purpose of the IAM systems on a cloud is to keep a unified and secure system for authorizing user access to cloud resources.

The least privilege principle does not imply that cloud resource access needs to be approved by the cloud service provider (CSP). Specifying the access rights to cloud resources for users is not the responsibility of the CSP, like Google, but of the company that is using the cloud. As per the shared responsibility model, the CSP is responsible for the security of the cloud, and the customer is responsible for security in the cloud, which includes managing access rights for users.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide](#)

Related Material

Question #54 of 60

Question ID: 1495255

Your company offers data analytics services using resources on its on-premises systems as well as its cloud deployment. It is now looking for a way to host its APIs both on-premises and on Google Cloud. Which Google Cloud service should you recommend for this?

- A) Apigee hybrid

✓
- B) Dataproc

✗
- C) Cloud Backup and DR

✗
- D) Cloud Trace

✗

Explanation

You should recommend Apigee hybrid for this scenario. With Apigee hybrid, you can place gateways near the source of application programming interface (API) traffic to use existing governance and security infrastructure while also benefiting from cloud-native services for monitoring and analytics. This is made possible by managing and controlling the runtime environment using Apigee hybrid. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Cloud Backup and Disaster Recovery (DR) in this scenario. Cloud Backup and DR is a service that provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensures low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #55 of 60

Question ID: 1495943

A digital imaging company using Google Cloud has asked you to calculate downtime for its image processing service. The service has an availability SLO of 99.5% for a month. What is the error budget of the service?

- A) 99%

✗
- B) 0%

✗
- C) 100%

✗
- D) 0.5%

✓

Explanation

The error budget in this scenario is 0.5%. This is equivalent to about three and a half hours of downtime. The error budget is the measure of the system’s possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%:

Error budget = 100% – SLO = 100% – 99.5% = 0.5%

When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

99% is not the error budget for this scenario. The value of 99% can be acceptable as the service level agreement (SLA) value. You need to keep an SLO value that is higher than the SLA, which helps you get early warnings when there is a danger of not meeting the SLA. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA which can cause issues for the company.

100% is not the error budget for this scenario. The value of 100% can be the desired value for reliability for a system, but it is not achievable in a real-world scenario. Costs and reliability need to be balanced in all real-world applications, and a system always depends on other systems, which reduces the level of reliability because the failure of one system can have an effect on others.

0% is not the error budget for this scenario. There is always a measurable value for an error budget because no system can have 100% uptime, and the error budget specifies the possible downtime that the system can face. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #56 of 60

Question ID: 1495197

A company that develops database applications is looking for a Google Cloud service that will enable its applications to get better performance by incorporating the availability and scalability of Redis services without the associated complexities of managing the Redis deployments. Which service should you use in this scenario?

- A) Datastore

B) Cloud Bigtable

C) Memorystore

D) Firestore
- ✖

✖

✔

✖

Explanation

You should use Memorystore in this scenario. Memorystore for Redis is a scalable, fully managed service for Redis on Google Cloud that provides 99.9% availability using up to five read replicas that are distributed across multiple availability zones (AZs). Memorystore provides automation for complex Redis and Memcached tasks, which include ensuring high availability, creating failovers, and performing patching. It is compatible with the protocols of both Memcached and Redis. Memorystore also has provision for private IP addresses and virtual private cloud networks (VPCs) for data security and has integration with Identity and Access Management (IAM) for access control. Memcached provides in-memory caching for temporary data storage. Redis is a key-value NoSQL database using an in-memory data system for high performance. Redis supports sharding and uses persistent data, while Memcached does not.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the Internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Dataprep is not the appropriate choice for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Cloud Bigtable is not the appropriate choice for this scenario. Cloud Bigtable is a NoSQL database service that is fully managed and enables you to work with massive workloads for analytics and operations with availability at 99.999%. It is ideal for uses that include applications for IoT, finance, banking, personalized services, and advertising technology. Cloud Bigtable has a storage engine that allows for improved predictions in applications utilizing ML. It also has integration with tools for big data including Hadoop, Dataproc, and Dataflow.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Memorystore](#)

Related Material

Question #57 of 60

Question ID: 1495245

Your company is looking for a serverless platform that is fully managed for running containers individually. What is the most efficient way to write code for an application and deploy it as a container with no additional overhead required?

- A) Compute Engine

B) App Engine

C) Cloud Functions

D) Cloud Run
- ✖

✖

✖

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Explanation

You should use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. The apps containerized with Cloud Run can be written in Ruby, Python, Go, Java, Node.js, or .NET.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides virtual machine (VM) computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google’s technological infrastructure.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Cloud Run](#)

Related Material

Question #58 of 60

Question ID: 1495938

Your company is looking into the security implications of moving its data to the Google Cloud Platform. As per the shared responsibility model, which of these are the responsibility of Google? (Choose two.)

A) Encryption keys

B) Access rights to Cloud Storage objects

C) Patching the guest operating system on a VM

D) Security of servers running Compute Engine instances

E) Data center security

✖

✖

✖

✔

✔

Explanation

Data center security and the security of servers running Compute Engine instances are both the responsibility of Google. The cloud service provider (CSP) takes on the responsibility of the cloud infrastructure, which includes the data center and its servers.

Access rights to Cloud Storage objects, encryption keys, and patching the guest operating system on a virtual machine (VM) are all related to data stored on the cloud, which means they are the customer's responsibility.

It is important to note that in the shared responsibility model, the customer is responsible for the security within the cloud, and the CSP is responsible for the security of the cloud. Google (the CSP) is responsible for the security of the cloud, including the data center, physical servers, and services. The security of the Google customer's workload is the responsibility of the customer. To summarize, the following are the responsibility of Google:

- Data center security
- Virtualization infrastructure
- Software and hardware infrastructure
- Networking infrastructure
- Global infrastructure, including Availability Zones, Regions, and Edge locations

The Google customer is responsible for securing all the data that they create and store in the cloud, including:

- Patching and configuring operating systems running on Compute Engine instances
- Setting up security groups
- Administering user accounts and permissions

However, the responsibilities can also depend on the Google service that is being used. For example, when using Cloud Spanner, Google is responsible for providing automated failover, redundancy, backups, and disaster recovery. Google also performs automated patching of the database engine. Similarly, the customer is responsible for the security of the code that is run using Cloud Functions. The customer is also responsible for patching software on Compute Engine instances.

When using Google Kubernetes Engine (GKE), the customer is responsible for workloads that contain docker files, application code, container images, Identity and Access Management (IAM) and Role-Based Access Control (RBAC) policies, and pods. The operating system in use on nodes, which can be Ubuntu or Container-Optimized OS (COS), is the responsibility of Google. GKE can automatically deploy upgrades to these images. This is separate from the OS that is running inside of containers.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Shared responsibilities and shared fate on Google Cloud](#)

Related Material

Question #59 of 60

Question ID: 1495932

A company wants to find a way to find anomalies in shopping transactions of users that browse its site and mobile app through a secure API-based system that is fully managed and scalable. Which Google Cloud service should you use for this?

A) Dataproc

B) Firestore

C) Payment Gateway

D) Looker

✖

✖

✔

✖

Explanation

You should use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing across multiple regions of Google Clouds global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Looker for this scenario. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts to proactively send timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Payment Gateway](#)

Related Material

Question #60 of 60

Question ID: 1497139

Your company uses various CRM systems for user experience management. It needs an intelligent system that can extract value from user data to better understand their needs and sentiments. Which Google technology can you use for this requirement?

- A) Cloud Translation

B) Looker

C) Memorystore

D) Contact Center AI
- ✖

✖

✖

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Explanation

Contact Center Artificial Intelligence (CCAI) is the appropriate choice for this scenario. CCAI is used to enhance the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow Customer Experience Edition (CX), Dialogflow Essentials (ES), Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights to extract topics, key areas, and sentiments from data using advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. Using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Looker is not the appropriate choice for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker creates alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Memorystore is not the appropriate choice for this scenario. Memorystore is a fully managed service that provides in-memory storage of data for providing fast data access with less than a millisecond of response time. It does this by the deployment of distributed caches for applications. It helps to migrate the cache layer to the cloud without needing to modify existing application code.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Contact Center AI](#)

Related Material